

MODULE HANDBOOK

CST3144

Full Stack Development

Faculty: Dr. Chinnu Mary George

***Department: Department of Computer Engineering
and Informatics***

Module Leader: *Dr. Mark Jones*

Semester 1 2026/2027

Credits: 30 Credits

Duration of the module 14 weeks

Intake: ***September***

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Note: This handbook is available in a large print format for students with any disability. If you would like a large print copy or have other requirements for the handbook, please contact CampusCentral via [AskMDX](#). We can supply sections from this publication as:

- a Word document with enlarged type — sent by email
- printed copy with enlarged type
- printed copy on non-white paper

Other formats may be possible. We will do our best to respond promptly. To help us, please be as specific as you can about the information you require and include details of your disability.

Welcome to Middlesex!



Module Coordinator: Dr Chinnu Mary George

Email: c.m.george@mdx.ac.ae

This module aims to develop a deep understanding of the latest full stack programming techniques, frameworks, and methodologies used by industry to develop the next generation software, which can be deployed on a wide range of devices and systems. The students will investigate, develop, and deploy the latest programming language standards, which are fundamental to app development, and currently being widely employed in industry. Modern programming frameworks will be introduced to provide the essential software architecture for large-scale software development, and the ability to target a wide range of platforms. The module will cover the three most important components of a complete architecture for a software application: Front-End, Back-End, and System Administration.

This Module Handbook introduces the module, its learning outcomes and requirements, and should be used alongside the Programme Handbook and the information available to all students on MyLearning: <https://mymdx.mdx.ac.uk/campusm/home#menu>

Note: Students must also refer to the [Campus Guide](#) on the Middlesex University Dubai website, which sets out key information relating to campus services, facilities, student support, policies, procedures, and behavioural expectations. It should be read in conjunction with this Module Handbook, the Programme Handbook, the [University Regulations](#), and the information available on MyLearning and MyMDX. Together, these documents set out the academic, administrative, and regulatory requirements applicable to your studies and form part of the framework governing your rights, responsibilities, and conduct as a student. Students are expected to familiarise themselves with, and comply with, the requirements and policies contained within these documents.

MyMDX App and Web Portal

The MyMDX App and Web Portal provide students with access to essential information and resources, including timetables, attendance records, campus news and events, live updates, student support services through [AskMDX](#), and Middlesex University Dubai social media channels. Access is available using the student's **@studentmdx.ac** credentials.

The web portal can be accessed at <https://mdxcentral.mdx.ac.ae>, while the mobile app can be downloaded from the Google Play Store and Apple App Store by searching for **MDX Dubai**. User guides for the [Student Portal](#) and [mobile app](#) are available for further guidance on using these platforms.

Your Module Teaching Team

Staff Name	Photo	University Contact Details	Advice & Feedback Hours	Role Description
Dr. Chinnu Mary George		Email: c.m.george@mdx.ac.ae Telephone: 04 367 8100	Tuesday 12:30am to 1:30pm	<i>Module Coordinator</i>
Mohamed Aatif		Email: m.shafi@mdx.ac.ae Telephone: 04 367 8100	Wednesday 1:00pm to 2:30pm	<i>Module Lab Tutor</i>

Communication with the Team

You are welcome to reserve an appointment during my office hours for timings that might suit us both. Please email me your appointment request along with your inquiry from your university email account. Don't forget to state your student ID number and module code and name in your email/online booking request. If you turn up at the door unannounced or phone at an unsuitable time, then you are far less likely to make contact.

I will send urgent messages about the module to you by email and/or the Microsoft Teams platform, so it is important that you read your university email and check your MS Teams notifications regularly. I recommend that you check your email and MS Teams accounts at least three times a week. I will also use these accounts to tell you about events (guest lectures or academic enrichment sessions, for example) and career opportunities (employability workshops, internships, job offers and so on) that will help enhance your learning.

Use your Middlesex email address when contacting members of staff to avoid emails being caught in University spam filters. Please also always include your full name,

student number, programme name, year of study and module details when contacting staff.

Students are required to use their Middlesex email address when contacting members of staff to avoid emails being caught in University spam filters. Students are also expected to always include their full name, student number, programme name, year of study, and module details when contacting staff.

Students are expected to respond to emails from staff within three working days (i.e. excluding weekends, public holidays, and University closure days) and allow staff the same period of time (three working days) to respond to their queries. Whilst staff may occasionally choose to respond to emails outside of normal working hours (Monday to Friday, 9 AM to 5 PM GST), this should not be expected as standard. If students do not hear back from a member of staff within the three working day timeframe, sending a reminder email is encouraged, but not sooner. Students may also raise their query with a different member of staff, such as their Campus Programme Leader/Coordinator. Contact details for these members of staff can be found within the MDX Central App. If a staff member is away from work, they may have set up an "out of office" automated email that provides instructions on how queries can be addressed in their absence; therefore, students are expected to carefully read any such messages.

Students are expected to contact their Module Leader/Coordinator(s) and/or the Campus Programme Leader/Coordinator if they are absent for any period of time and are unable to attend scheduled teaching sessions.

Note: Students are expected to check the Middlesex University Dubai website, MyMDX App and MDX Dubai App regularly during the semester for any other notifications or announcements. Attempt to look for basic answers to questions (e.g. by using MyMDX, Dubai Campus Guide or MDX Dubai App) before contacting lecturers or staff.

For all queries that don't relate to your programme of study (e.g. fees, wellbeing, accommodation, IT issues, etc.) you should directly contact Campus Central via [AskMDX](#) on the MDX Dubai App rather than your lecturers. This will help ensure you get a response more quickly.

Take time to write polite emails/ queries (as you would in professional employment) in all communications with staff. This should include use of a clear subject line to indicate the subject of your message and previous emails on the conversation thread.

1. Module Overview

1.1 Aims

This module aims to develop a deep understanding of the latest full stack programming techniques, frameworks, and methodologies used by industry to develop the next generation software, which can be deployed on a wide range of devices and systems. The students will investigate, develop, and deploy the latest programming language standards, which are fundamental to app development, and currently being widely employed in industry. Modern programming frameworks will be introduced to provide the essential software architecture for large-scale software development, and the ability to target a wide range of platforms. The module will cover the three most important components of a complete architecture for a software application: Front-End, Back-End, and System Administration

1.2 Learning Outcomes

LO Number	Description	Attribute
LO1	Justify and deploy the latest programming standards, technologies, and strategies required for advanced software development.	Knowledge and Understanding
LO2	Evaluate the main characteristics, strengths, and weaknesses of the latest software architecture frameworks.	Knowledge and Understanding
LO3	Critically evaluate the methodologies of developing platform-independent software, and the strength and weakness of existing solutions.	Knowledge and Understanding
LO4	Create a Front-End solution by employing best practices for advanced software development.	Skills
LO5	Deploy and manage essential technologies for dependency management.	Skills
LO6	Design and develop a Back-End solution with advanced technologies including software hosting and data storage.	Skills

1.3 Syllabus

- Programming Front-End solutions.
- Advanced interactivity for Front-End solutions.
- Version Control for software systems.

- Refactoring of software systems.
- Advanced Design and Development of software applications.
- Full Stack Development and Dependency Management.
- Advanced Back-End technologies and Administration.
- Programming Platform-Independent software applications.
- Communication between Front-End and Back-End technologies.
- Deployment of Software Applications.
- App Hosting and Data Storage.

1.4 Learning and Teaching

Weekly contact hours:

- Workshop: 2 hrs (activities including for example Theory, Practice, Tutorials, CW Demo, CW Support, CW QCA, etc., depending on the stage of the Module)
- Laboratory: 3 hrs
- Drop-in online session: 1 hr

Short key concept videos will be made available.

The focus of teaching will be a mixture of activities, lab-based practical work, and discussions, using a Project-Based Learning (PBL) approach. Skills and experience are built up progressively through weekly lab sessions in order that the students can implement a full stack software application. Various tools for software development will be introduced throughout the module as they become necessary. The student will develop their individual application through in-class discussions, peer-to-peer collaboration with other students, and self-study.

Total hours:

Scheduled teaching: 72 hours

Independent study: 228 hours

1.5 Learning Planner

Date & Learning week	Lecture*	Lab	Teaching Contributor(s)	Mapped Learning Outcome(s) and Assessment(s)
September 14, 2026, Week 0	Introduction to the module & handbook	Concept checking and review	CG	LO1
September 21, 2026 , Week 1	Programming Front-End Solutions	Programming Front-End Solutions	CG	LO1
September 28, 2026, Week 2	Advanced Interactivity for Front-End Solutions	Advanced Interactivity for Front-End Solutions	CG	LO1
October 5, 2026 , Week 3	Version Control of Software Systems and Refactoring of Software Systems	Version Control of Software Systems and Refactoring of Software Systems	CG	LO2
October 12, 2026 , Week 4	Advanced Design and Development of Software Applications	Advanced Design and Development of Software Applications	CG	LO2
October 19, 2026 – Week 5	Full Stack Development and Dependency Management	Full Stack Development and Dependency Management	CG	LO2
October 26, 2026 – Week 6	Advanced Back-End Technologies and Administration	Advanced Back-End Technologies and Administration	CG	LO3
November 2, 2026, Week 7	Programming Platform-Independent Software Applications	Programming Platform-Independent Software Applications	CG	LO3
November 9, 2026, Week 8	Communications between Front-End and Back-End Technologies	Communications between Front-End and Back-End Technologies	CG	LO4

Date & Learning week	Lecture*	Lab	Teaching Contributor(s)	Mapped Learning Outcome(s) and Assessment(s)
November 16, 2026 , Week 9	Deployment of Software Applications	Deployment of Software Applications	CG	LO4
November 23, 2026 , Week 10	App Hosting and Data Storage; Review and Coursework Time	App Hosting and Data Storage; Review and Coursework Time	CG	LO5
November 30, 2026 , Week 11	Review and Coursework Time	Review and Coursework Time	CG	LO6
December 7, 2026 – Week 12	Review and Coursework Time	Review and Coursework Time	CG	LO6
December 14, 2026 – Week 13	CW Demo	CW Demo	CG	Submission :Dec 14th 2026

1.6 Assessment Scheme

(a) Formative assessment scheme

Students will receive formative feedback on summative assessments tasks. Students will be expected to work towards a complete software application, mastering related architecture and technologies. The students will be formatively assessed by using a Project-Based Learning (PBL) approach, which will support students in the labs.

(b) Summative assessment scheme

The portfolio gives students the opportunity to put into practice the theories, frameworks, and libraries covered in the module. The portfolio will focus on the Full-Stack aspects of a complete software application development.

Task: Full Stack Software Application Development (Individual) This application development portfolio will allow students to apply knowledge and skills in the latest programming standard to design and implement a full-stack software application architecture.

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The deliverable of this portfolio will consist of software code and students are expected to run a short demonstration of the application.

Weighting	Specification e.g. word count / duration / no. of pages	LO mapped to	Anonymous ly marked	Ethics approval required
100%	Demonstration (maximum 10 minutes). Zip file of the project (code of the resulting software) must be no more than 10MB. Deadline: week 11.	1- 6	☒ No ☐ Yes	☒ No ☐ Yes – <i>individual student</i> ☐ Yes – <i>group approval</i> ☐ Yes – <i>whole module</i>

In order to pass the module, the student will be required to achieve either:

- ☒ an overall aggregate of grade 16;
- ☐ an overall aggregate of grade 16 with a minimum of grade 16 in each assessment component
- ☐ an overall aggregate of grade 16 with a minimum of grade 17 in each assessment component
- ☐ an overall aggregate of grade 16 with a minimum of grade of 18 in each assessment component

1.7 Learning Resources

This module has a variety of learning resources available for you to use to support your learning. These include recorded lecture, lecture slides, feedback, and key reading materials. These can be accessed online via the module page. Please visit the module page regularly to make use of these.

1.8 Research Ethics

The teaching, learning, assessment and research activities undertaken in this module have been considered and are not likely to require ethical approval.

However, please seek advice if undertaking the module entails carrying out any research activities involving human participants, human data, animals/animal products, precious artefacts, materials or data systems or social media. If you submit work that includes data gathered from or about people, this may be treated as academic misconduct and could lead to fail grade being awarded.

Research ethics approval seeks to ensure all research is designed and undertaken according to certain principles of ethical research. These include:

1. Primary concern must be given to the **safety, welfare and dignity** of participants, researchers, colleagues, the environment and the wider community
2. Consideration of **risks** should be undertaken before research commences with the aim of minimising risks to those involved – i.e. human participants or animal subjects, colleagues, the environment and the wider community, as well as actual or potential risks to those directly or indirectly affected by the research.
3. **Informed consent** should be freely given by participants, and only by a trained person when collecting or analysing human tissue.

4. Respect for the **privacy, confidentiality and anonymity** of participants
5. Consideration of the rights of **people who may be vulnerable** (by virtue of perceived or actual differences in their age, social status, ethnic origin, gender, mental capacities, or other such characteristics) who may be less competent or able to refuse to give consent to participate
6. For participants below the age of 18, parental consent must be obtained prior to the study.
7. Researchers have a responsibility to the general public and to their profession; as such they should balance the anticipated benefits of their research against **potential harm, misuse or abuse** which must be avoided
8. Researchers must demonstrate the highest standards of **ethical conduct and research integrity**. They must work within the limits of their skills, training and experience, and refrain from exploitation, dishonesty, plagiarism, infringement of intellectual property rights and the fabrication of research results. They should declare any actual or potential conflicts of interest, and where necessary take steps to resolve them.
9. When using human tissues for research, the UK's Human Tissue Act and Human Tissue Authority (HTA) requirements must be met. Please contact the relevant designated person (DP) in your department or the HTA Designated Individual (DI).
10. Research should **not involve any illegal activity**, and researchers must comply with all relevant laws

You can apply for research ethical approval using the Middlesex University Dubai Ethics Form. The relevant forms, templates and guidance on the approval process can be obtained from the module folder on MyMDX.

Your module coordinator can provide further guidance. Additionally, documents can be found through the following link: <https://bit.ly/35Zp7MZ>

2. Assessment

2.1 Formative Assessment

Formative assessments help show you and us that you are learning and understanding the material covered in this course and allow us to monitor your progress towards achieving the learning outcomes for the module. Although formative assessments do not directly contribute to the overall module mark, they do provide an important opportunity to receive feedback on your learning.

In this module, students will receive formative feedback on summative assessments tasks. Students will be expected to work towards a complete software application, mastering related architecture and technologies. The students will be formatively assessed by using a Project-Based Learning (PBL) approach, which will support students in the labs.

2.2 Summative Assessment

Summative assessment is used to check the level of learning on the course. It is summative because it is based on accumulated learning during the course. The point is to ensure that students have met the

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learning outcomes for the course and are at the appropriate level. It is the summative assessment that determines the grade that you are awarded for the module.

There is 1 assessment component in this module:

- Coursework 1
 - o Task: Full Stack Software Application Development (Individual) This application development coursework will allow students to apply knowledge and skills in the latest programming standard to design and implement a full- stack software application architecture.
 - o The deliverable of this coursework will consist of software code and students are expected to run a short demonstration of the application.

Summative assessment	Weighting	Draft Feedback opportunity	Op-Deadline	Feedback
Coursework 1	100%	Week 10	Coursework demo (Week 11 & 12)	Within 15 working days since student's demonstration

In order to pass this module, you need to pass the assessment with a minimum grade of 40% or equivalent.

Reassessment for this module normally takes place in the following way:

If students fail the module (fail grades are 17, 18, 19, 20 with an overall mark between 0% and 39%) they are eligible for a re-sit. Students will be re-examined in the assessed component(s), which they have failed (please note that this can be a different assessment task). Information on what element to re-sit will be made available on the module's MyMDX site approximately two weeks after the module results have been published on MyMDX. You will be notified about reassessment dates in a timely manner.

Before submitting your work, ensure all sources are accurately referenced and that spelling, punctuation, and grammar are correct. Written assessments evaluate technical proficiency in English; details are outlined in the assessment marking criteria and the [University Grade Criteria Guide](#).

Reasonable adjustments will be made for those students who have a declared disability/specific learning condition which would affect performance in this area.

Reassessment for this module normally takes place in the following way:

2.3 Assessment Brief

Assessment Title	Coursework 1
Assessment Type	Software Development Project and Demonstration

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Description	You will create a fully functional web app for a fictitious company, using the techniques covered in this module. The front-end, back-end and data will all be hosted online. A full description of the requirements for the web app will be made available later in the semester. The deliverable for this assessment will consist of a zip file containing the code for the web app as well as links to the online front-end, back-end and database. Students will also need to give a short demonstration where they answer questions about the project. Demonstrations will be scheduled during teaching hours at the end of semester.
Submission date, time	Monday 14th December at 11pm
Feedback type G date	Verbal feedback during demonstration; written feedback with 15 working days of the demonstration
File size	The submitted zip file must be no more than 10MB in size.

Assessed learning outcome(s) G evidence	Students will demonstrate all learning requirements through the successful development of the web app, and through their answers to questions about their code and the technologies used.
Assessment weighting %	100%
Submission Method	Links to submit the zip file and book a demonstration will be provided on the module's MyLearning page.
Assessment Criteria	1. Submission Instructions Your CW submission will consist of a single zip file. The zip file must be no more than 10MB in size, and it must contain the following elements: • your code in 2 folders respectively related to: ○ your Vue.js App. ○ your Express.js App (do not include the 'node_modules' folder, otherwise the zip will be too big to submit). • A README file with the following links: ○ [Vue.js App] the link to your GitHub Repository. ○ [Vue.js App] the link to your GitHub Pages from where the app can directly run. ○ [Express.js App] the link to your GitHub Repository. ○ [Express.js App] the link to the render.com route that returns all the lessons. • All collections (including the 'lesson' and 'order' collections) exported from your MongoDB Atlas. See here for how to export collection in MongoDB Compass : https://docs.mongodb.com/compass/current/import-export#export-data-from-a-collection • the requests you created in Postman. See here for how to export requests in Postman: https://learning.postman.com/docs/getting-started/importing-and-exporting/exporting-data Instructions for the text area when uploading your submission: - please include the same information as requested for the README file. Overall Requirements and Instructions • You can use an external CSS library such as Bootstrap. Make sure the library file or online link is included in the submission. A submission could receive zero marks if it fails any of the following requirements:

- [GitHub Repositories] t
 - The code of the Vue.js App must be hosted in a GitHub repository, with at least 10 commits, and the code of the Express.js App must be hosted in another GitHub repository, with at least 10 commits.
- [Front-End]
 - the App must be implemented by using Vue.js framework. Other frameworks or related technologies (e.g., React, AngularJS, Svelte, Apache, XAMPP) are NOT allowed.
 - Any JavaScript library is NOT allowed if it duplicates or replaces features provided by Vue.js framework. Check with the tutor if not sure.
- [Back-End and Connection details for the Front-End]
 - The Back-End server must use “Node.js”; others such as Apache or Xampp are not allowed.
 - The REST API must be developed with “Express.js”.
 - The Front-End data access must be achieved with “promise” using “fetch” function; “XMLHttpRequest” or library such as axios.js are not allowed.
 - The connection to MongoDB (in your Express.js App) must use the native Node.js driver only; libraries like Mongoose are not allowed.
- [Hosting]
 - The Front-End Vue.js app must be hosted and demonstrated on GitHub Pages.
 - The Back-End server must be hosted on render.com. Other hosting services are not allowed. The address of your hosted server must include your student number, e.g.
<https://M001234567-after-school.onrender.com>

The data must be stored in MongoDB Atlas and retrieved via your Express.js App; local MongoDB or any other databases are not allowed.

Use of GenAI Tools:	Software Code: AMBER – Assistive scope GenAI tools may be used to help you write sections of your code more efficiently. However they are not a substitute for understanding and being able to edit your code. Note in particular the following: • GenAI should only be used to create short sections of your code. Using GenAI to develop your whole project in one go is expressly prohibited. You are responsible for making sure all the parts of your code work together correctly. • If you use GenAI tools to generate the code for you without understanding the technologies or without being able to describe your code properly during demonstration, then you will get a low mark for this Module. • Any use of GenAI must be declared as part of your submission (see the table “Record and Acknowledgement of GenAI Use” in the module handbook. Demonstration: RED (no scope) The use of GenAI during the demonstration is prohibited.
Middlesex Graduate Competencies	☐ Leadership and Influence ☒ Curiosity and Learning ☐ Communication, Empathy and Inclusion ☐ Collaborative Innovation ☐ Entrepreneurial Mindset ☒ Problem Solving and Delivery ☐ Resilience and Adaptability ☒ Technological Agility

For assessment criteria, please refer to Appendix 1 at the end of this document.

2.4 Academic Integrity and Misconduct

Academic integrity means producing your work in a transparent way, based on valid academic and professional sources, properly referenced and with any assistance fully acknowledged.

Note: Compliance with the [Academic Integrity and Misconduct Policy](#) is a condition of enrolment. Students are responsible for ensuring that they are fully aware of, and adhere to, the requirements set out in the Policy.

Academic Misconduct is a breach of the values of academic integrity, and can occur when a student cheats in an assessment, or attempts to deliberately mislead an examiner that the work presented is their own when it is not. It includes, but is not limited to, plagiarism, self-plagiarism, commissioning or buying work from a third party or copying the work of others, breach of examination rules, collusion etc.

- **Plagiarism** – using extensive unacknowledged quotations from, or direct copying of, another person's work and presenting it for assessment as if it were your own effort.
- **Purchasing or Commissioning** - This includes the use of third party essay writing services. Essay mills are now illegal entities and use of them is facilitating an illegal activity.
- **Collusion** – working together with other students (without the tutor's permission) and presenting similar or identical work for assessment.
- **Self-Plagiarism** – including any material which is identical or substantially similar to material that has already been submitted by you for another assessment in the University or elsewhere.
- **Unauthorised use of Generative Artificial Intelligence (GenAI)** – using GenAI without appropriate acknowledgement and referencing. Appropriate use of GenAI is detailed the [AI in Learning and Teaching Policy](#).

To support your academic integrity, please refer to the [Referencing guidance](#) and ensure your referencing meets your discipline's required standards.

Our [Library](#) and the [Centre for Academic Success \(CAS\)](#) run workshops and clinics to help you learn how to avoid plagiarism and how to reference correctly. To get support and guidance on academic writing techniques that meet our expectations of Academic Integrity, please contact a staff member in [CAS](#).

Student Success Essentials (an online self-study course available via MyLearning) includes useful information about how to approach your assessments and complete them with honesty. The course also describes what plagiarism (cheating) is and how to avoid it, so you don't face any action due to violation of academic integrity rules.

The Dubai Academic Integrity and Misconduct Team (aim@mdx.ac.ae) provide guidance on responding to allegations of academic misconduct and its related processes.

2.5 Appropriate use of GenAI

This table defines the different levels of scope you may be given to use GenAI in your assessed work. Your Assessment Brief above defines which level applies to that assessment task and will include further guidance and support.

GenAI Scope in Assessment	What it means
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RED – No scope	GenAI is not permitted in the development or creation of outputs (except assistive technology via reasonable adjustment). Assessments in this category can include exams, Vivas/interviews, and professional / regulatory competencies where GenAI is out of scope.
AMBER – Assistive scope	Assistive use is permitted only within defined parameters for assessment preparation and process, such as: summarising sources, connecting ideas/themes, self-test and revision prompts, spelling and grammar support in drafts, support for data analysis, project planning and management. GenAI should not be used to produce entire sections of text or elements of media that appear in the final submission.
GREEN– Integrated Scope	Integrated use: GenAI can be used to create content that appears in the final submission. GenAI can be used at multiple stages of the assessment process. GenAI use must be acknowledged and/or cited, and assessments may ask for student commentary and critical evaluation of its use.

2.6 Record and Acknowledgement of GenAI Use

Use this table to record how you have used GenAI for this assessment. It will help you keep track of the tools and prompts you used, reflect on your learning, and explain your study choices if needed - for example in tutorials, group work, reflections, or reflective and methodological sections of your work.

	Details and Explanation
GenAI tools used	<i>e.g. MS Copilot, ChatGPT, Midjourney, Perplexity, etc.</i>
Date of use	<i>e.g. 15/07/2026</i>
Amber or Green Category & Purpose of use	<i>e.g. Amber – spell checking; Green – to suggest an outline essay structure.</i>

Examples of prompts used (keep a log of all your prompts saved)	<i>e.g. "Generate ideas for structuring an essay on the themes of..."</i>
How did you check the validity of the GenAI suggestions?	<i>e.g. Cross-checked against reading list materials; Google Scholar; discussion with tutor.</i>
The section or phase of assessment where GenAI was used	<i>e.g. Planning project work; finding themes in data; producing an outline structure; checking draft sections for grammar; integrating and evaluating GenAI output in final work.</i>

2.7 Micro-course: Foundations of AI Literacy

Students are expected to complete the micro-course on 'Foundations of AI Literacy' on MyLearning to empower yourself with the basic skills you will need to navigate the rapidly changing landscape of generative artificial intelligence. This micro course has lessons that will equip you with essential skills to understand AI, evaluate AI generated content to make informed judgement about tools you encounter and use AI responsibly as part of your student journey. This micro course can be completed within 90 minutes. You can expect to learn through:

- Completely free and accessible bite-sized content based on the latest guidelines from thought leaders at UNESCO, IBM, Google, OpenAI, and Anthropic
- Core AI literacy concepts written by leading academics and our information literacy experts in the library
- Progressing at your own pace using self-study materials and interactive content, including videos, case-studies, quizzes, and interactive activities
- Upon completing the micro-course, you will be awarded a certificate of completion.

What does this micro-course cover?

Unit 1. Understand: AI and ethical aspects: This unit introduces the fundamental differences between AI and Generative AI, emphasizing the role of data in AI applications. It also addresses key ethical concerns related to AI-generated content.

Unit 2. Evaluate: Judgement and responsible use: This unit focuses on evaluating AI-generated output to identify bias and hallucinations, while developing the skills required to use AI responsibly. It also provides guidance on evaluating AI tools from an ethical perspective in academic contexts.

Unit 3. Use: Prompting skills and use of tools: This unit explores frameworks and approaches to help you develop effective prompting skills. It also examines the type of tasks best suited to popular AI tools and their applications in academic and research contexts.

Students can register on this course via the [Library](#).

2.8 Support that is available to you

Support and draft feedback sessions for Coursework 1

Additional support

Please use this space to specify any additional support students can utilise. This could refer to:

- *Lecturer Consultation: can be requested by prior appointment by sending an email to x.yyyy@mdx.ac.ae or provide a booking link*
- *MyMDX: the folder on MyMDX relevant to this assessment task includes the TurnItIn online submission link, sample assignments and template format.*
- *CiteThemRight: offers online help for referencing at: <http://www.citethemright-online.com.ezproxy.mdx.ac.uk>*
- *Centre for Academic Success (CAS) Consultation: you have the option to book individual consultations or group appointments with members of the CAS Team. These sessions have been known to help students reach their academic goals through one-on-one consultations with our highly professional and helpful CAS team members. For more information on the services offered by the CAS Team, or to book an individual consultation, you may send an email to CAS@mdx.ac.ae or visit <https://www.mdx.ac.ae/cas>*
- *CAS specific sessions (in case you have arranged CAS to be involved in your module)*
- *Library specific sessions (in case you have arranged the Library to be involved in your module)*
- *Instructional Resources: The Library offers many instructional resources via their website: <https://www.mdx.ac.ae/library> where students can learn how to use the Library and its resources by watching videos, using interactive tools, and exploring Library Guides – which provides students with the information necessary to understand how to access books, eBooks, Journals, articles, databases, conference proceedings, research results, etc.*
- *Liaison Librarians – Students are encouraged to contact their Liaison Librarian who can provide expert individual or group assistance finding materials for specific assignments, give research support and direction, and provide general Library guidance: <https://www.mdx.ac.ae/library/liaison-librarians>*

- *An Introduction to the Research Process: This online course is specifically designed to introduce Middlesex University Dubai students to the five steps of the Research Process. The course can assist with getting started on research for an essay, term paper, presentation, or dissertation by providing guidance on locating Library and web resources including books, articles and eBooks. <https://www.mdx.ac.ae/library/an-introduction-to-the-research-process>*
- *Library Focus: Students are encouraged to register for Library Focus workshops. Library Focus workshops develop research skills and help students best utilise Library resources*
- *Further teaching and learning support materials can be found on the Dubai Teaching and Learning Support MyMDX page that includes short videos, tutorials, etc produced by the CAS and Library teams. <https://mdx.mrooms.net/course/view.php?id=24990>*

Assessment support for students of determination

Students who have declared special needs or learning differences or an ongoing medical condition may contact the Inclusion Counsellor in the Centre for Academic Success. This will set out the reasonable additional support that will be put in place by the University to assist that student's learning and assessment. It is the student's responsibility to make arrangements and follow the procedures set by the Centre of Academic Success in order for reasonable adjustments to be put in place. This may include the option to take up to 5 calendar days late submission on individual written coursework without having a grade reduction. Check the Middlesex University's [Policy and Procedures for Extenuating Circumstances Claims 2025-2026](#) for more information.

2.9 Feedback on your assignments

Students receive helpful and informative feedback on all assessments to support their learning and development. Feedback may be provided in various forms, including written comments, tutorial feedback (individual or group), peer feedback, or other appropriate methods. Formative or draft assessments will receive feedback but no grade, and comments will highlight strengths as well as areas for improvement. All assignments should be submitted online unless stated otherwise in the assessment brief.

Feedback on summative assessments will normally be provided within 15 working days of the published submission date.

3. Results Confirmation

3.1 How is your assignment mark agreed?

The following diagram provides an overview of the marking process for your module assessment:

Step 1	You submit your assignment
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Step 2	The first marker grades the work* and provides feedback; this could be completed anonymously depending on the assessment type.
Step 3	A moderator or second marker reviews a sample of the work to ensure the consistency of the grades and feedback. A final mark for the work is agreed between the first marker and the moderator or second marker.
Step 4	A sample of work, from a selection of modules across the programme, is reviewed by the External Examiner to check that the grading and feedback is at the right level and in line with external subject benchmarks (this applies to levels 5,6 &7 only).
Step 5	Your final grades are submitted to the Programme assessment board.

***Provisional Grades:** At the end of your first semester, you can see your module grades in the Grades and Progress tile within MyMDX. These grades are provisional and not yet confirmed by the Programme Assessment Board.

Final Grades and Progression: After your second semester, the Programme Assessment Board will confirm your grades. Then, your final module results, progression status, or finalist classification will be posted in the Grades and Progress tile within MyMDX.

Each assessment component will be marked in percentages. However, the overall module grade will be calculated using the University 20-point scale:

20-point scale	General scale	General scale (full ranges)	Percentage used for aggregation purposes only (for areas marking directly to the 20 point scale on modules with multiple assessment components)
1	80% - 100%	79.50% - 100%	90%
2	76% - 79%	75.50% - 79.49%	77.5%
3	73% - 75%	72.50% - 75.49%	74%
4	70% - 72%	69.50% - 72.49%	71%
5	67% - 69%	66.50% - 69.49%	68%
6	65% - 66%	64.50% - 66.49%	65.5%
7	62% - 64%	61.50% - 64.49%	63%
8	60% - 61%	59.50% - 61.49%	60.5%

9	57% - 59%	56.50% - 59.49%	58%
10	55% - 56%	54.50% - 56.49%	55.5%
11	52% - 54%	51.50% - 54.49%	53%
12	50% - 51%	49.50% - 51.49%	50.5%
13	47% - 49%	46.50% - 49.49%	48%
14	45% - 46%	44.50% - 46.49%	45.5%
15	42% - 44%	41.50% - 44.49%	43%
16	40% - 41%	39.50% - 41.49%	40.5%
17	35% - 39%	34.50% - 39.49%	37%
18	30% - 34%	29.50% - 34.49%	32%
19	0% - 29%	0.01% - 29.49%	15%
20	Non-participation	0%	0% (non-submission of a component)

For more information visit the [MyMDX Assessment FAQ page](#) or reach out to Campus Central via [AskMDX](#).

3.2 Extenuating Circumstances

Students must submit each component of assessment by the deadline set by the Module Leader/Coordinator. If circumstances beyond your control affect your ability to meet an assessment deadline or perform well, these may be considered extenuating circumstances (ECs). If you find yourself in this situation, there are academic tools available to you to help you manage your studies. Please visit the [MyMDX Extenuating Circumstances and Late Submission](#) page for more information.

For individual written coursework only, late submission of up to 24 hours from the deadline is permissible, however, the grade for the component is reduced by 10% or equivalent (or less where this would reduce a pass grade below 40%). Students should take into consideration the impact this will have on their deadline schedule and their final grades before selecting this option.

Extenuating Circumstances can only be requested for summative assessment, not for formative assessment. For example, only for assessment that counts towards your overall module grade.

Students **MUST** provide evidence/supporting statement with any request for extenuating circumstances to be considered (except in cases of self-certification*) for a deferral to the next assessment opportunity, or for Extenuating Circumstances to be noted at the Programme Assessment board.

***Self-certification** can only be used when an extenuating circumstance **has affected you for period of 7 days or less**, i.e. a short illness that occurs at the point of assessment submission. You can apply for the following outcomes due to your Extenuating Circumstances. Please note the outcome of any Extenuating Circumstances application may be different from what has been requested:

Deferral	Your circumstances have impacted on your ability to sit your exam or complete and/or submit your assessment. You are requesting an opportunity to sit the exam or submit the work at the next assessment opportunity
ECs noted for the board only	Your circumstances may have affected your performance on other assessments for one or more modules and, although you have sat the exam or submitted the assessments, you wish the Programme Assessment Board to be aware of this in case there is a borderline decision to be made regarding your progression/finalist decision.
Deferral & ECs noted for the Board	Your circumstances have impacted on your ability to sit your exam or complete and/or submit your assessment. You are requesting an opportunity to sit the exam or submit the work at the next assessment opportunity.

Note: There could be progression (i.e. additional year(s) of study), financial (tuition fees and scholarships) and student visa implications as a result of deferring your assessment.

For more information on how to apply for Extenuating Circumstances, please check the dedicated [webpage](#) or contact CampusCentral via [AskMDX](#).

4. Expectations of Studying

Students are expected to actively participate in both online and on-campus learning, behave professionally, and engage respectfully, professionally and in a supportive manner with peers, tutors, and staff. They should come prepared for classes and communicate courteously in all University-related interactions, including email.

Students must familiarise themselves with the [Campus Guide](#) and [University Regulations](#), which outline key policies, services, procedures, rights, responsibilities, and standards of conduct. Compliance with these requirements is expected throughout their time at Middlesex University Dubai and for the duration of their enrolment.

4.1 Attendance and Engagement

Regulations regarding attendance can be found on the [University Regulations](#). In accordance with University Regulation for taught programmes of study, it is the responsibility of students to attend scheduled classes and prescribed activities for the modules on which they are registered.

Students should attend and engage with all scheduled classes and prescribed activities. Studies have shown that a good student engagement has a positive impact on performance and therefore is an important factor in helping you to fulfil your academic potential. In addition, for those who are on student visas, Dubai's regulatory authorities require attendance to be monitored.

Lecturers maintain attendance records during scheduled teaching sessions using the MDX Dubai App. Students are expected to follow any guidelines and instructions provided for the proper recording of attendance during learning sessions.

The MDX Dubai App (available on iOS and Android) allows students to register attendance at timetabled classes. To record attendance:

1. Connect to the internet using the **#mdxDUBAI** Wi-Fi network.
2. Open the MDX Dubai App.
3. Log in using a Campus User ID (M0xxxxxxx@studentmdx.ac) and password.
4. Ensure that permission has been granted for the MDX Dubai App to access the smartphone's location and camera settings.
5. Locate the correct module and timetabled class via the Calendar.
6. When the lecturer or tutor displays the class QR code, use the scanner within the MDX App to scan the code.
7. An automated notification will appear within the App confirming that attendance has been recorded.

Middlesex University Dubai supports students in achieving their full potential through a range of strategies designed to provide a supportive learning environment. Online

support material on MyMDX is provided as a guide to class content but is no substitute for interaction with tutors and classmates. Students experiencing difficulties beyond their control that prevent engagement with a module should notify their tutor and Campus Central, who may be able to offer support and guidance.

Note: Where your attendance and engagement fail to meet the minimum levels required (normally **75%** of scheduled learning sessions and activities but could be higher) to attain the learning outcomes of the module, you may be excluded from the assessment. You may have the opportunity to retake the whole module with permission from the Campus Programme Coordinator/ Leader, without grade penalty, though you will have to pay the relevant additional tuition fees for the module re-registration.

4.2 Recording Lectures

As per the [University Regulations](#) unauthorised audio recording, video recording or photography of lectures, or other forms of learning activities by students, is prohibited.

Limited recording by students may be permitted under exceptional circumstances only (for example, for an individual student as a “reasonable adjustment”, within the meaning of the [UK’s Equalities Act](#)), upon explicit permission provided by the tutor and, where appropriate, by everyone else involved. Permission for recording does not imply permission for publication (e.g. Instagram, YouTube, or other Social Media), or distribution to others. Unauthorised recording of such activities violates the privacy of persons involved, may infringe on copyrights and intellectual property rights of others and can be intrusive and disruptive in a learning environment. In all cases, violation of this regulation will be managed under the student disciplinary procedures.

In addition to a violation of [University Regulations](#), unauthorised recordings may expose students to other unintended consequences, as per UAE law. The United Arab Emirates has several laws (for example, [Federal Decree Law No. 34 of 2021 on Combatting Rumours and Cybercrimes](#)) for the protection of privacy and reputation and defamation. Some of the acts that could amount to a criminal offence are:

- possessing on an electronic device a photo taken without the subject’s consent
- posting other people’s pictures or videos online or on social media (including WhatsApp) without their consent
- tagging a person without their consent
- threatening or insulting people online
- spreading information via social media, that is not verified by the official sources
- gossiping about people or maligning them.

4.3 Laptops, Mobile Phones and Other Devices

Technology can enhance learning in higher education when used responsibly. Students are encouraged to use laptops, tablets, mobile phones, and other devices for

learning activities, and tutors may require them in some sessions. When not being used for class activities, devices should be on silent or airplane mode. Calls, texts, and social media should be avoided unless approved by the tutor. Disruptive use of devices may result in removal from class and disciplinary action.

4.4 Professional Behaviour

Students must come to sessions prepared and ready to contribute where appropriate.

University Student ID cards must be carried at all times while on campus, and students must be able to identify themselves when requested. Students must also comply with community health precautions and other health and safety protocols.

Communication with fellow students, tutors, and University staff, including email correspondence, must be conducted in a professional, formal, and courteous manner.

Unacceptable Behaviours in Face-to-Face Interactions

- Aggressive or abusive behaviour, including shouting, bullying behaviour, physical violence, rudeness, and making threats, inappropriate gestures, or indecent comments
- Persistently interrupting or disrupting learning activities and events
- Engaging in antisocial behaviour that impacts others
- Making derogatory or discriminatory remarks about others
- Using offensive language or engaging in personal verbal attacks
- Discussing sensitive matters in public settings
- Making unsupported claims that the University or individual staff have committed criminal, corrupt, biased or perverse conduct without any evidence
- Demanding that staff set aside or make exceptions to University regulations
- Demanding responses within an unreasonable timescale or insisting on seeing specific staff members when not feasible
- Refusing to accept outcomes or solutions offered or repeatedly seeking further explanations without new information or evidence

Actions to be taken by the University or Staff

- Highlight the unacceptable behaviour(s) and explain why and what effect it is having
- Pause or end the discussion or interaction if the unacceptable behaviour remains persistent
- In serious cases, ask for removal of individuals from University facilities and report the incident to relevant Dubai authorities
- Restrict personal contact and require communication via email or written channels and/or through third parties
- Referral to appropriate disciplinary procedures

Version Control

Disclaimer: The material in this document is as accurate as possible at the date of production; however, you will be informed of any major changes in a timely manner.

Version	Date	Author	Updates	Approved by
1.0	02.09.2026	Institutional Planning and Quality Office (IPQO)	Added Dubai Specific information	Senior Officer
2.0	07.09.2026	Senior Management Team and IPQO	Refinement	Deputy Director

Appendix 1: Marking Criteria

Marking Criteria

Core Functionality (60%)

It is expected that most students will be able to complete all these tasks.

- **[Front-End] “Display List of Lessons” functionality (7%):**
 - A. there should be at least 10 lessons, and each lesson should have 5 spaces (or availability) (1%).
 - B. each lesson should have at least (5%): Subject (1%), Location (1%), Price (1%), Spaces (or availability: this indicates how many spaces are left) (1%), a Font Awesome icon (or an Image) (1%).
 - C. v-for must be used for the display of the lesson list (1%).
- **[Front-End] Sort functionality (8%):**
 - A. the user can choose to sort the lessons by one of the following attributes: subject, location, price, or spaces (i.e. availability) (5%).
 - B. there must be an option to sort in ascending or descending order (order dependent on the sorting attribute selected), which should work for each of the attributes (3%).
- **[Front-End] “Add to Cart” functionality (5%):**
 - A. each lesson must have an “Add to Cart” button (1%).
 - B. the button is always visible, and only enabled when space is larger than 0 (1%).
 - C. clicking the button once (related interactions implemented by using v-on) will add one space to the shopping cart, reducing the remaining space by one (2%).
 - D. once there is no more space, i.e. space = 0, the “Add to cart” button should be disabled but still visible, i.e. clicking it will not further reduce “space” nor add lessons to the cart (1%).

- **[Front-End] “Shopping Cart” functionality (5%):**
 - A. the shopping cart button should only be enabled after at least one lesson is added to cart (1%).
 - B. clicking the shopping cart button should show the cart page, and clicking the button again goes back to the lesson page (1%).
 - C. the shopping cart, in the cart page, should show all the lessons added (1%).
 - D. in the shopping cart page, the user should be able to remove lessons from the shopping cart; the removed lesson is added back to the lesson list (in the lesson page) (2%).
- **[Front-End] Checkout functionality (6%):**
 - A. the checkout is part of the shopping cart page (not part of the lessons page) (1%).
 - B. the “checkout” button is always visible and only enabled (clickable) after valid “Name” and “Phone” are provided (2%).
 - C. the “Name” must be letters only and the “Phone” must be numbers only; the check must be done using JavaScript (suggestion: regular expressions) (2%).
 - D. clicking the “checkout” button should display a message confirming the order has been submitted (1%).
- **[Back-End] MongoDB should have (8%):**
 - A. a collection for lesson information (minimal fields: topic, price, location, and space) (4%).
 - B. a collection for order information (minimal fields: name, phone number, lesson IDs, and number of space) (4%).
Suggestion: the element lessonIDs can contain 1 or more lesson IDs, depending on how many different kinds of lessons you have in your order. Other solutions could be accepted: in fact, how you design this is not the primary aspect of this module, therefore it is up to you to find a reasonable solution that works satisfactorily.
- **[Back-End] Middleware Functions implemented in the Express.js Server should include (8%):**
 - A. a “logger” middleware that outputs all requests to the server console; the student needs to be able to inspect and explain the log (4%).
 - B. a static file middleware that returns lesson images, or an error message if the image file does not exist; the student needs to be able to test and demonstrate these aspects (4%).

- **[Back-End] REST API implemented in the Express.js Server should include and the following (you will also be asked to test and explain the routes during the demonstration) (7%):**

- A. one GET route /lessons, which returns all the lessons as a Json (2%).

Example:

```
[  
  { 'topic': 'math', 'location': 'Hendon', 'price': 100, 'space': 5 },  
  { 'topic': 'math', 'location': 'Colindale', 'price': 80, 'space': 2 },  
  { 'topic': 'math', 'location': 'Brent Cross', 'price': 90, 'space': 6 },  
  { 'topic': 'math', 'location': 'Golders Green', 'price': 95, 'space': 7 },  
]
```

- B. one POST route, which saves a new order to the “order” collection (2%).
- C. one PUT route, which can update any attribute in a lesson of the “lesson” collection (suggestion/requirement: this will be used, after an order is submitted, for updating the available spaces: updating to any number the available spaces, and not just increasing or decreasing) (3% if the requirement is fully covered, 1% if the solution is just increase or decrease).

- **[Front-End] Fetch Functions implemented in the Front-End should include (6%):**

- A. one fetch that retrieves all the lessons with GET (2%).
- B. one fetch that saves a new order with POST after it is submitted (2%).
- C. one fetch that updates the available lesson space with PUT after an order is submitted (2%).

Challenge tasks (20%)

These tasks are not covered in detail in the lectures and labs, so you will have to do some research and thinking to complete them. It is not expected that everyone will be able to complete these tasks.

You **must** be prepared to explain your solution and how it works, describing how you implemented your solution in the front-end, middleware, server, and/or database (as applicable). A solution for this part of the coursework could receive **zero** if you are not able to explain it.

These tasks should be implemented using the techniques taught in the module, without using any additional libraries.

- **[Challenge] Search Functionality (6%)**

- The user can search for a lesson, without specifying which attribute to search on. For example, searching for “a” should return all the lessons with “a” in its “title” or “location” or “price” or “availability”.
- The search should be implemented in the back-end and the front-end should receive the results via REST API.
- This task is worth at most 3% if the search is implemented only in the front-end (i.e. by filtering the results returned from the /lessons route).

- **[Challenge] Teachers (8%)**

- The database also stores information about the teachers for each lesson. Each lesson has a first name, a last name, and is associated with one or more lessons that they teach. The user can choose a teacher and view all the lessons taught by that teacher.

- **[Challenge] Enrolment (6%)**

- Implement a GET route /enrolment/:lessonid which takes as input the id of a lesson, and returns the enrolment number for that lesson – i.e. the number of slots that have already been purchased.

Demonstration (20%)

During your demonstration session, you will show your completed web app and the code for your project and explain how it works.

Remember that the demonstration is a **mandatory** part of the assessment – students who do not attend the demonstration will receive zero for the whole submission.

In order to ensure that the demonstration runs smoothly, please make sure you have the following open on your device before your allotted start time:

- The front-end of your app on Github Pages
- The logs for your server on render.com
- Your database on MongoDB Atlas
- The source code for your front-end and back-end in Visual Studio Code (or another IDE)
- Your API requests in Postman

During the demonstration, you will be asked to do the following:

- **Demonstrate Front-End functionality (5%):**
 - Show the front-end of your website behaving as intended i.e. displaying lessons, adding lessons to cart, placing orders, etc.
- **Inspect server log (3%):**
 - Show the console log output in render.com from your ‘logger’ middleware and explain what the logs mean. (1% if the student is able to show the log output but not able to explain it)

- **Demonstrate requests in Postman (5%)**

- For each of the following REST API routes, you should have a Postman request prepared in advance and be ready to demonstrate it:
- A. the GET route /lessons, which returns all the lessons as a Json (1%).
- B. the POST route, which saves a new order to the “order” collection (2%).
- C. the PUT route, which can update any attribute in a lesson of the “lesson” collection (2%).

- **Answer questions about your code (7%)**

- During the demonstration we will ask 5-10 short questions about different parts of your code and their function. Make sure you understand your code so that you are able to answer these questions correctly.

- **Explain your solutions to “challenge” tasks:**

- Explain your solutions to any challenge tasks you complete, showing in the code how you implemented your solution in the front-end and back-end.

(See above for the marks associated with each challenge task)